ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Section A: objective questions (30) · Section B: microcontroller architecture, communication protocols, operational amplifiers, DAC and ADC, motor drivers, power supplies and batteries (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The memory in a microcontroller that stores the program and keeps it when power is removed is …

A SRAM. B a register only.
C cache. D flash memory.

1.1.2 Which communication protocol uses only two wires, SDA and SCL, to connect many devices?

A SPI B I²C C UART D USB

1.1.3 SPI communication uses the lines MOSI, MISO, SCK and …

A SDA. B TX only.
C SS (chip select). D RX only.

1.1.4 The voltage gain of a non-inverting amplifier with R_f = 47 kΩ and R_i = 4,7 kΩ is …

A 11. B 10. C −10. D 0,1.

1.1.5 An ideal operational amplifier has …

A zero gain. B infinite input impedance and very high open-loop gain.
C low input impedance. D high output impedance.

1.1.6 An 8-bit DAC with a 5 V reference has a step size of about …

A 4,88 mV. B 0,5 V. C 1 V. D 19,53 mV.

1.1.7 A hardware signal that makes the microcontroller stop its main program and run a special routine immediately is a/an …

A interrupt. B loop. C variable. D library.

1.1.8 A 7805 voltage regulator gives an output of …

A 7,8 V. B 12 V. C 5 V. D 3,3 V.

1.1.9 A 3-cell (3S) lithium-polymer battery has a nominal voltage of …

A 3,7 V. B 9 V. C 11,1 V. D 12,6 V.

1.1.10 The integrated circuit L298N is commonly used as a …

A dual H-bridge motor driver. B voltage reference.
C memory chip. D Wi-Fi module.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 UART communication needs a shared clock line.

1.2.2 Each device on an I²C bus has its own address.

1.2.3 An inverting amplifier's output is 180° out of phase with its input.

1.2.4 A voltage follower (buffer) has a gain of 1.

1.2.5 A linear voltage regulator wastes the excess voltage as heat.

1.2.6 A switching (buck) converter is usually less efficient than a linear regulator.

1.2.7 LiPo batteries can be safely discharged to 0 V per cell.

1.2.8 Battery capacity is measured in ampere-hours (Ah) or milliampere-hours (mAh).

1.2.9 A comparator is an op-amp used without feedback to compare two voltages.

1.2.10 Increasing the bit resolution of an ADC makes its step size larger.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Part of the microcontroller that performs arithmetic and logic operations	A	logic level shifter
1.3.2	Holds the address of the next instruction	B	stall current
1.3.3	Non-volatile memory used to store settings that can be changed by the program	C	program counter
1.3.4	Resets the microcontroller if the program stops responding	D	DAC
1.3.5	Number of bits transferred per second in serial communication	E	watchdog timer
1.3.6	Device that converts digital values into an analogue voltage	F	ALU
1.3.7	Op-amp circuit whose output is the sum of several inputs	G	summing amplifier

1.3.8	Lets a 3,3 V device communicate safely with a 5 V device	H	baud rate
1.3.9	Small board that monitors and balances LiPo cells	I	battery management system
1.3.10	Current that flows into a motor when it is prevented from turning	J	EEPROM

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: MICROCONTROLLERS AND COMMUNICATION

2.1 Draw a block diagram of a microcontroller showing the CPU, flash memory, RAM, input/output ports, timers and the ADC, connected by a bus. (6)

2.2 Explain the difference between a microcontroller and a microprocessor. Give an example of each. (4)

2.3 Compare UART, I²C and SPI under the headings number of wires, speed and typical robot device. (9)

2.4 Explain why an interrupt is better than polling for detecting pulses from a wheel encoder. (3)

[22]

QUESTION 3: OPERATIONAL AMPLIFIERS AND DATA CONVERSION

3.1 A force sensor gives 0 to 0,3 V and must be amplified to 0 to 3,3 V for a microcontroller.

3.1.1 Calculate the gain needed. (2)

3.1.2 Use a non-inverting amplifier with R_i = 1 kΩ. Calculate R_f. (3)

3.1.3 Draw the circuit diagram of the non-inverting amplifier. (4)

3.2 An inverting amplifier has R_f = 100 kΩ and R_i = 20 kΩ. Calculate the output voltage for an input of 0,5 V. (3)

3.3 An 8-bit DAC with a 5 V reference receives the binary value 1000 0000. Calculate the output voltage. (3)

[15]

QUESTION 4: MOTOR DRIVERS

4.1 Draw an H-bridge using four switches (or transistors) and a motor, and explain which switches are closed for forward and for reverse. (7)

4.2 Explain what would happen if both switches on the same side of the H-bridge were closed at the same time, and how motor driver ICs prevent this. (3)

4.3 A robot motor has a stall current of 2,5 A. The motor driver can deliver 2 A continuously. Explain the risk and suggest TWO solutions. (3)

[13]

QUESTION 5: POWER SUPPLIES AND BATTERIES

5.1 A 7805 regulator supplies 0,5 A to the electronics of a robot from a 9 V battery.

5.1.1 Calculate the power wasted in the regulator. (3)

5.1.2 Explain why a heat sink may be needed and suggest a more efficient alternative. (3)

5.2 A robot uses a 2 200 mAh battery and draws an average current of 550 mA. Calculate the running time. Why will the real running time usually be less? (4)

5.3 Give FIVE safety rules for charging and storing lithium-polymer (LiPo) batteries. (5)

5.4 Explain why the grounds (0 V) of the motor supply and the logic supply must be connected together, while noisy motor currents should not flow through the logic wiring. (5)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER